

Plan: regular- N -gon prism family — unify the triangle, add
pentagon + hexagon

HyMeKo manipulation program (Track A2, shapes)

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1 Scope and goal

Generalize the one-off triangular-prism branch into a **parametric regular- N -gon prism** generator, and add **O7-P** (pentagon, $n = 5$) and **O8-H** (hexagon, $n = 6$) to the curriculum. The corner count n becomes a shape axis: $n = 3$ (triangle) $\rightarrow 5 \rightarrow 6 \rightarrow \dots \rightarrow \infty$ (circle). Chosen by the user over per-shape branches precisely because it is the §6.5-clean move: one generator, no per-polygon `elif` growth.

Why unify (§6.1/§6.5). `compose_planar_scene` already special-cases "triangle"; adding pentagon/hexagon as more `elif` branches would be a Cartesian per-shape dispatch. Instead a single `_regular_prism_vertices(n_sides, R, half)` plus one equal-area circumradius helper serve all n . The triangle becomes the $n = 3$ special case.

2 Key invariant — triangle bit-parity

The equal-area circumradius of a regular n -gon (area = $\frac{1}{2}nR^2 \sin \frac{2\pi}{n}$ set equal to πr^2) is

$$R_{\text{eq}}(r, n) = \sqrt{\frac{2\pi r^2}{n \sin(2\pi/n)}}.$$

For $n = 3$ this equals the existing triangle expression $\sqrt{\pi r^2 / (\frac{3\sqrt{3}}{4})}$ **bit-identically** (verified: $|\Delta| < 10^{-15}$). The vertex generator uses the same apex-up angle set $\{\pi/2 + k 2\pi/n\}$ and the same extrusion loop order, so for $n = 3$ the emitted MJCF mesh string is **byte-identical** to today's. $\Rightarrow$ the committed O6-T result is unchanged; a parity test enforces this.

Footprints (equal area to the coin, $r = 0.02$): $n=3 \rightarrow 31.10$ mm, $n=5 \rightarrow 22.99$ mm, $n=6 \rightarrow 21.99$ mm, $\rightarrow 20$ mm as $n \rightarrow \infty$.

3 Affected files (full list)

- `hymeko_rl/env/object_spec.py`: add `Shape.NGON = "ngon"` (keep `TRIANGLE` as a backward-compat alias); add `n_sides: int | None` field + `__post_init__` precondition (`NGON  $\Rightarrow$   $n\_sides \geq 3$`); add `polygon_sides()` (3 for `TRIANGLE`, `n_sides` for `NGON`, else `None`); add a pure module helper `equal_area_regular_ngon_circumradius(r, n)` (single source of truth); generalize `footprint_radius`; thread `n_sides` through `compose_kwargs/planar_env_kwargs/from_`

- `hymeko_rl/env/planar_grasp_env.py`: rename `generalize_tri_prism_vertices` → `_regular_prism_vertices` (`R, half_z`); add a `disk_n_sides` kwarg; route `coin_shape` ∈ {triangle, ngon} through the general generator using the shared circumradius helper (imported from `object_spec`; no import cycle — verified).
- **NEW** `data/robotics/galambos_env_o7_pentagon.hymeko, ..._o8_hexagon.hymeko`: @disk shape "ngon"; `n_sides` 5|6; radius 0.02.
- `hymeko_rl/coin_delivery/object_curriculum.py`: add O7-P, O8-H.
- `hymeko_rl/tests/test_object_spec.py, hymeko_rl/tests/test_object_curriculum.py`: parity + intent + membership.
- **NEW** `reports/2026-08-12-ngon-prism-family/`, this plan dir.

4 CORE.YAML items touched

None. `object_spec.py` and `planar_grasp_env.py` are `hymeko_rl/**` (Python), not core (core = the Rust crates, `docs/spec/*`, `rtl/*`, pinned deps). `compose_planar_scene` is the *sanctioned* builder (the ownership guard forbids only *new callers*, not edits to the builder itself). No dependency added.

5 Interface changes

Additive. `ObjectSpec` gains an optional `n_sides` field (default `None` ⇒ every existing spec unchanged). `compose_planar_scene` gains an optional `disk_n_sides` kwarg (default `None` ⇒ existing calls unchanged). `Shape` gains `NGON`; `TRIANGLE` is retained. No existing signature is broken.

6 Test strategy (by layer)

- **Parity (regression)**: the emitted triangle mesh vertex string via the generalized generator is byte-identical to a frozen reference (the $n = 3$ path); the compiled O6-T mass + geom-type are unchanged. This would fail if the refactor altered the triangle.
- **Unit (object_spec)**: `Shape.from_str("ngon")`; `NGON n_sides` precondition (missing / < 3 asserts); `polygon_sides()`; `footprint_radius` for $n = 5, 6$ equals R_{eq} ; the shared helper reduces to the triangle at $n = 3$.
- **Unit (object_curriculum)**: membership incl. O7-P/O8-H; pentagon/hexagon geom `MESH`, mass ≈ O0, footprint between the coin and the triangle.
- **Integration**: generation smoke over the full 8-variant curriculum — 0 model/contract failures, O0–O6 certified-capture counts *unchanged* vs the prior run (regression), O7/O8 generate + certify straddle.
- **Static**: `ruff, mypy --strict` on changed non-test code; the ownership-guard tests stay green.
- **Performance**: peak RSS < 2 GB; record wall.

7 Performance budget

Peak RSS < 2 GB (hard cap 16 GB). Wall: 8-variant smoke ≈ 30 min CPU (36 → 48 rollouts; the O6-T run was 36 rollouts / 21 min). No GPU. Seeds fixed.

8 Risk anticipation

- **Triangle regression** — highest risk (edits the frozen builder path). Mitigated by the bit-parity test + the smoke regression (O0–O6 counts must match). If parity breaks: halt, do not proceed.
- **Mass parity for $n = 5, 6$:** MuJoCo density-derives mass from the convex mesh; the intent test asserts $\approx$ O0 within 10^{-4} . If off, the object is not a clean shape-only ablation.
- **Convexity / degenerate mesh:** regular polygons are convex for all $n \geq 3$; no risk. A `n_sides < 3` precondition rejects degenerate input at the boundary.
- **Import direction:** `planar_grasp_env` $\rightarrow$ `object_spec` for the circumradius helper; verified no cycle (`object_spec` is a leaf).
- **Worst-case input:** full 8-variant curriculum $\times$ positions $\times$ seeds through the smoke; no training. Within budget.

9 Rollback path

Single commit. Revert restores `_tri_prism_vertices`, drops `Shape.NGON / n_sides` / the two scenes / the two curriculum entries. No state mutation, no checkpoints.

10 Out of scope

No teacher-bank / retrieval / RL / delivery. Generation + capture characterization only, as with O6-T. The round family (ellipse / capsule) is a *separate* later plan.